OpenFIRE Light Gun & Wireless Dongle User Manual

Firmware: OpenFIRE-Firmware-ESP32 (ESP32-S3 light gun + ESP-NOW Dongle, English OLED UI)

Optional configuration app: OpenFIRE Desktop App (Team OpenFIRE / OpenFIRE-App-cn)

Document version: 1.0 | Date: June 2026

Note: OLED text in this manual matches the default English firmware strings (build without -D OLED_MENU_ZH).

1. Product Overview

OpenFIRE is a four-point infrared (4IR) light gun system. The ESP32-S3 port supports two PC connection methods: (1) USB wired direct connection; (2) ESP-NOW wireless via a Dongle. In wireless mode the PC sees the same HID device as wired—no extra game or force-feedback adapters are required.

This manual covers: light gun (GUN) daily use, OLED pause menu, calibration, player IDs (P1–P4); wireless receiver (Dongle) and wireless pedal pairing, on-screen status, and troubleshooting.

1.1 Package contents (reference)

- Light gun (IR camera, buttons, OLED, optional rumble/solenoid, etc.)
- USB cable (wired play, charging, or firmware flashing)
- Wireless Dongle (ESP32-S3, plugs into PC USB; SSD1306/TFT models show pairing status on OLED)
- IR emitter bar(s) (mount on monitor top/bottom for Square layout, or on all four sides for Diamond)
- (Optional) Wireless pedal module for Time Crisis–style cover mechanics

1.2 Safety

- On boot and during auto stick center calibration (~2 s), do not touch the analog stick.
- Solenoid recoil produces mechanical impact and heat—never point the gun at people or animals.
- Monitor the temperature readout on the OLED during long sessions; overtemperature protection may limit FFB.
- Flash firmware or rewire only when powered off. See docs/BOARDS_PIN.md for board pin maps.

2. First Boot & Connection

2.1 Boot sequence

After power-on the gun runs this sequence automatically:

Stage	OLED text (English)	Meaning
Stick auto-cal	CALIBRATION READY	~2 s—do not move stick
USB detection	go in usb mode → wait for usb mount	Wired vs wireless path
Link ready	!! LINK READY !! / USB mounted	USB or wireless link up
First-time aim cal	Welcome! / Pull trigger / start calibration!	Pull trigger to start 5-point cal

2.2 Wired USB

- Connect the gun to the PC with USB.
- Wait for Windows to enumerate the HID device (name typically FIRECon P1–P4 depending on Gun ID).
- Top bar may show USB mounted ; lower-left shows USB icon.
- Use the OpenFIRE Desktop App on the COM port for advanced settings (Docked mode—see Section 10).

2.3 Wireless Dongle (recommended order)

- Step 1: Plug the Dongle into the PC USB port. Wait for startup (OLED: Scanning WiFi / best channel eval / Please wait, ~10–15 s).
- Step 2: Do NOT plug the gun into PC USB. Power on the gun.
- Step 3: When the Dongle shows Ch:xx Pair / Turn on lightgun / ESP-NOW listen, the gun should start pairing broadcast.
- Gun OLED top bar may show Ch:xx Waiting link.
- Step 4: After success the Dongle shows RF link OK → Player / Channel / Gun MAC; PC shows the gun's configured device name.
- Step 5: With a wireless pedal, the Dongle may show AWAIT PEDAL; the gun shows Search Pedal [xxs] while searching.
- Step 6: Wi-Fi icon at lower-left indicates wireless mode.
- On subsequent boots the gun prefers the last paired Dongle; if that fails it rescans all channels.

2.4 Boy Mode (offline play without PC)

If wireless is down and USB is not mounted, hold the trigger within ~3 s of power-on to enter Boy Mode: OLED shows Boy? BOOT/TRIG Xs countdown, then BOY MODE READY . Allows local solenoid/rumble testing without a PC (hardware dependent).

3. OLED Status Bar (Gameplay Screen)

Area	OLED example	Meaning

Top left	P1: P_A Diam 42C	Player ID, profile slot (P_A…P_D), layout Diam/Squa, temperature
Top right	OFF / 300	Play Timer: OFF = disabled; number = remaining seconds
Full top row	Prof: ProfileA P1	Shown when pausing or switching profiles
Lower left	USB / Wi-Fi icon	Wired or wireless Dongle
Lower center	RF / LOW / AF	Rumble FFB, Low Button mode, Autofire (abbreviations)
Lower right	Mouse / Gamepad / MiSTer icon	Current USB output mode

Transient top-bar messages include: Mode Changed: Gamepad, Autofire: ON, Low Button: OFF, Rumble FFB: ON, Sent Escape Key!, Play time over, etc.

4. Pause Menu (English OLED)

The pause menu switches profiles, runs calibration, changes USB mode, sets Gun ID, saves settings, etc. Two interaction styles exist, controlled by the simplePause option (default is usually Direct Pause hotkeys).

4.1 Enter / exit pause

Action	Button combo	OLED hint
Enter pause	Reload + Select	Using + current profile name
Exit pause	Reload + Home	Return to gameplay screen
Hold to pause (if Hold to Pause enabled)	Trigger + A ~2.5 s	—
Simple Pause hold exit	A + B ~half Hold to Pause time	—

4.2 Runtime shortcuts (no pause menu)

Function	Button combo	OLED message
Send Esc to PC	Reload + Start	Sent Escape Key!
Send Esc (quick)	Trigger + A + C (release all together)	Sent Escape Key!
Mouse/KB ↔ Gamepad	A + B + Trigger	Mode Changed: Mouse/KB / Gamepad

Note: Gun C maps to the Reload button bit. MiSTer mode is selected via Mode Change in Simple Pause.

4.3 Direct Pause (default hotkey mode)

In pause, four lines show A > / B > / Str> / Sel> with profile names; most actions use combos:

Function	Button combo
Screen calibration	Trigger while paused
Save all settings to Flash	Start + Select

IR sensitivity +	B + Up
IR sensitivity −	B + Down
Run mode Normal	Start + A
Run mode Average / Average2	Start + B
Send Esc	Reload + Start or Trigger+A+C
Mouse/KB ↔ Gamepad	A + B + Trigger
Switch profile A/B/Start/Select	Press matching face button on pause screen
Skip center step during cal	Press A during calibration

4.4 Simple Pause (scrolling list mode)

If Simple Pause is enabled in the OpenFIRE App: A = previous item, B = next item, Trigger = confirm. Menu labels match firmware exactly:

OLED label (English)	**Function**
Center Calibrate	Set current stick position as new center (requires analog pins)
Range Calibrate	~4 s stick rotation + release to center (RangeCal: rotate → Release to center → OK)
Calibrate	Enter 5-point screen aim calibration
Profile Select	Pick Profile A/B/Start/Select (sub-screen: A>/B>/Str>/Sel>)
Mode Change	Cycle Mouse/KB → Gamepad → MiSTer (Current: …)
Layout Toggle	Square ↔ Diamond (Layout: Square / Diamond)
Gun ID / Gun ID (P1-P4)	Set player P1–P4; updates USB PID and device name
Stick Output Mode	Stick: Gamepad / D-Pad / Keyboard Arrows
Deadzone	Stick deadzone 0–30% (Deadzone: N%)
Axis Mode	Signed / Unsigned axis
Swap Sticks	Swap IR aim axis with physical stick L/R
Play Timer	0 / 5 / 10 / 15 / 20 min (Play Timer: OFF / Play Timer: N min); expiry: Play time over
Save Gun Settings	Write profiles & globals to Flash (Saving... / Save successful)
Rumble Toggle	Master rumble motor (Toggling Rumble ON/OFF)
Solenoid Toggle	Master solenoid (Toggling Solenoid ON/OFF)
Autofire Toggle	Autofire (Autofire: ON/OFF)
Low Button Toggle	Low-button mapping (Low Button: ON/OFF)
Rumble FFB Toggle	Use rumble instead of solenoid for serial FFB (Rumble FFB: ON/OFF/N/A)
Analog Invert X / Y	Invert stick axis
Send Escape Keypress	Send one Esc key to PC

Items without hardware are skipped automatically. Gun ID sub-menu: A/B pick P1–P4, Trigger saves, Reload+Home cancels.

5. Gun ID & Multiple Guns (P1–P4)

- Each gun can be P1, P2, P3, or P4 (USB Product ID 1–4, device name e.g. FIRECon P1).
- Set via OLED Gun ID: Pn , Simple Pause Gun ID , or App → Settings → USB Identity.
- After changing Gun ID you must Save Gun Settings and replug USB for Windows to see the new ID.
- Multi-gun: assign unique IDs; use separate USB ports or one Dongle per gun.

6. USB Output Modes

Mode	OLED	Typical use
Mouse/KB (default)	Current: Mouse/KB	MAME, Demul, TeknoParrot, etc.
Gamepad	Current: Gamepad	Games expecting gamepad HID
MiSTer	Current: MiSTer	MiSTer FPGA platform

Switch via Simple Pause Mode Change , runtime A+B+Trigger (Mouse/KB ↔ Gamepad), or Desktop App.

7. Calibration Profiles

Four built-in profiles: Profile A, Profile B, Profile Start, Profile Select.

Idle bar shows P_A…P_D shorthand; profile picker may show Name (Square) or Name (Diamond) .

- Direct Pause: press the button matching A>/B>/Str>/Sel> on the pause screen.
- Simple Pause: Profile Select submenu.
- Desktop App: Profiles tab → Confirm to save.
- Always Save Gun Settings or Start+Select before power-off.

8. Calibration Guide

8.1 Screen aim calibration (required)

- Maps IR coordinates to your monitor. Required on first use or after changing displays.
- Start: pull trigger at first boot; pause Calibrate ; Direct Pause trigger; App cal window.
- Top bar during cal: Cali: + profile name.
- Steps: Top → Bottom → Left → Right → Center → Verify. Reload+Home cancels and restores old data.

8.2 Analog stick calibration

Type	OLED menu / message	Action
Boot auto center	CALIBRATION READY	~2 s at every boot—don't touch stick
Center Calibrate	Center calibrated	Set current position as center

Range Calibrate	RangeCal: rotate / OK	Rotate stick ~4 s, release to center

Stick range calibration is done on the gun OLED (Range Calibrate); the Desktop App has no range-cal UI yet.

8.3 IR emitter mounting

- Square layout (Squa): emitters on top and bottom of the monitor.
- Diamond layout (Diam): emitters at the mid-point of all four sides.
- Use the App's Open IR Emitter Alignment Assistant for a layout diagram.

9. Wireless Dongle

9.1 What the Dongle does

The Dongle is an ESP32-S3 receiver on the PC. It talks to the gun over ESP-NOW (2.4 GHz) and presents the same USB identity and forwards HID plus serial force-feedback data.

9.2 OLED status (English)

OLED text	**Meaning**	**What to do**
OpenFIRE DONGLE	Boot logo	Wait
RF sniff ch / Scanning WiFi	Picking least-congested channel	Leave USB plugged ~10–15 s
Best channel eval / Please wait	Channel survey	Wait
Ch:xx Pair / Turn on lightgun / ESP-NOW listen	Waiting for gun	Power gun (no PC USB on gun)
RF link OK / Wait gun stream	RF up, waiting HID stream	Wait a few seconds
AWAIT PEDAL	Waiting for wireless pedal	Power pedal if used; else wait
Player: N / Channel: N / Gun: MAC	Paired	Test on PC

TFT Dongles (e.g. LilyGO) may show Searching / best channel / Scanning WiFi / Connecting / AWAIT PEDAL / Player:N.

9.3 Wireless pedal

- The pedal has no OLED. While searching, the gun top bar shows Search Pedal [xxs].
- After the gun pairs, the Dongle may show AWAIT PEDAL until the pedal handshake completes.
- Pedal gameplay matches a wired pedal; status is LED-only on the pedal unit.

9.4 PC USB & COM port

- Before pairing completes the PC may see a placeholder device; afterward the gun's configured name appears.
- Force feedback (Mamehooker, etc.): select the Dongle COM port at 9600 baud.
- Change VID/PID/mode on the gun, then re-pair so the Dongle refreshes its USB descriptor.

9.5 Multi-device tips

- Recommended: one gun + one Dongle. Multiplayer: unique Gun IDs and separate Dongles.
- Avoid multiple Dongles in pairing mode competing for the same gun.
- If a gun sticks to an old Dongle MAC, clear binding or wait for timeout and full-channel rescan.

10. OpenFIRE Desktop App (optional)

- Pin mapping, toggles, profile parameters, USB P1–P4, IR calibration, button test, FFB test.
- Connect via COM port; App sends Dock command to enter Docked mode and sync data.
- Click Confirm after edits. Gun ID: Settings → USB Identity.
- Use a firmware build compatible with your App branch (OpenFIRE-App or OpenFIRE-App-cn).

11. Force Feedback & MAMEHOOKER

- With MAMEHOOKER enabled at compile time, gun/Dongle expose a 9600 baud serial channel.
- PC FFB software → serial → (wireless: via Dongle) → gun drives solenoid/rumble.
- OLED may show Mamehook ammo/life when configured.
- Rumble FFB Toggle chooses rumble motor vs solenoid for serial FFB (separate from Rumble Toggle).
- Solenoid Toggle = solenoid hardware; Rumble Toggle = rumble motor hardware.

12. Troubleshooting

Symptom	Likely cause	Fix
Dongle stuck on Turn on lightgun / Searching	Gun off, gun on PC USB, out of range	Dongle first, then gun; unplug gun USB; move closer
Gun keeps Search Pedal	Pedal off or out of range	Power pedal; move near Dongle
PC name stays P1	Gun ID not saved or USB not replugged	Save Gun Settings; replug USB
App cannot set Gun ID	App/firmware protocol mismatch	Update to matching firmware/App pair
Wireless drops after connect	2.4 GHz congestion	Reduce distance; fewer Wi-Fi devices
OLED blank	Wrong I2C address/pins for board	Verify PlatformIO board env; replug USB
Aim offset	Uncalibrated or new monitor	Re-run screen cal; check Diam/Squa vs emitter mount
FFB dead	Wrong COM or FFB disabled	Dongle COM @ 9600; enable Rumble FFB Toggle; check wiring
Locked after Play time over	Play Timer expired	Wait for lockout to end or set Play Timer: OFF
Stick drift on boot	Stick touched during cal	Don’t touch stick first 2 s; run Center/Range Calibrate

Appendix A: Quick daily workflow

[Wired] USB to PC → confirm USB icon → play.

[Wireless] Dongle to PC → gun on (no USB to gun) → Wi-Fi icon → play.

[Settings/cal] Reload+Select → pause → change → Save Gun Settings or Start+Select → Reload+Home.

[Esc] Reload+Start or Trigger+A+C.

[Multiplayer] Unique Gun ID per gun → save → replug USB / one Dongle each.

Appendix B: Chinese vs English firmware (menu cross-reference)

Chinese (OLED_MENU_ZH)	English (default)
中心 校准	Center Calibrate
摇杆 校准	Range Calibrate
屏幕 校准	Calibrate
配置选择	Profile Select
少键模式	Low Button Toggle
震动 FFB 开关	Rumble FFB Toggle
游戏计时	Play Timer
玩具模式	Boy Mode
枪号	Gun ID
类 同 ST / SC	Diam / Squa

Appendix C: Supported boards (ESP32-S3)

- ESP32-S3 WROOM-1 DevKitC-1 (N16R8 / N8R2)
- Waveshare ESP32-S3-Pico / ESP32-S3-Zero (N8R8 / N4R2)
- ESP32-S3 Super Mini (4 MB) and ESP32-S3 N8R8 (8 MB Flash + 8 MB PSRAM)
- See docs/BOARDS_PIN.md and lightgun/platformio.ini environments for build targets.

Appendix D: Resources & license

- Firmware: OpenFIRE-Firmware-ESP32 (github.com/wei134102/OpenFIRE-Firmware-ESP32)
- Upstream: Team OpenFIRE / alessandro-satanassi OpenFIRE-Firmware-ESP32
- Board pinouts: docs/BOARDS_PIN.md
- OpenFIRE is open source; ESP32 port, ESP-NOW wireless, and OLED i18n by community maintainers.
- Features depend on your flashed firmware version and hardware configuration.